

Tom Marty

 3rdcore.github.io

 tom.marty@mila.quebec

 Google Scholar

  

EDUCATION

Ph.D. in Machine Learning <i>Mila, Université de Montréal (Leave of absence) - GPA 4.00</i>	Jan. 2024 – Mar. 2026 <i>Montréal, Canada</i>
M.Sc. in Machine Learning <i>Polytechnique Montréal - GPA 3.91</i>	Sep. 2021 – Aug. 2023 <i>Montréal, Canada</i>
B.Sc. in Applied Mathematics <i>X 2018, Ecole Polytechnique - GPA 3.84</i>	Sep. 2018 – Aug. 2021 <i>Palaiseau, France</i>
Advanced Preparatory Class for Competitive Exams <i>Lycée Jean-Baptiste Say - GPA 4.00 - Top 0.1% national</i>	Sep. 2016 – Aug. 2018 <i>Paris, France</i>

RESEARCH INTEREST

- **Broad interest** : Representation Learning, Generative Modeling, AI for Science, Information Theory
- **Methodological interest** : Large-Scale Training, Diffusion model, Entropic compression
- **Applications** : ML for Biology, Robust ML, open-ended Decision Making, Robotics

INDUSTRY AND ACADEMIC EXPERIENCE

Visiting Researcher <i>ServiceNow Research</i>	Apr. 2023 – Sep. 2023 <i>Montréal, Canada</i>
• Developed WorkArena: an open-source Gym environment for evaluating Agent at solving common-knowledge tasks on a Web Browser	
Visiting Researcher <i>Corail Research Group</i>	Jan. 2021 – Sep. 2021 <i>Montréal, Canada</i>
• Developed SeaPearl : an open-source Constraint Programming solver with RL-based heuristics • Supervised five interns on the development of the open-source project SeaPearl (30K lines of code) • Implemented a DQN pipeline with heterogeneous GNNs to approximate optimal branching decision	
Software Engineer Intern <i>Dronisos, The drone light show company</i>	Jun. 2020 – Sep. 2020 <i>Bordeaux, France</i>
• Developed <i>Harmony</i>, a physics based meta-heuristic that secures massive 1000 drones swarms • <i>Harmony</i> reduced the allocated securing time in production from 2 weeks (handmade) to 0.2 seconds • Achieved automatic securing on the company first 1000 drones choreography (+500k\$ show)	

CONFERENCE AND JOURNAL PUBLICATIONS

A Compression Perspective on Simplicity Bias

Tom Marty*, Eric Elmoznino, Tejas Kasetty, Léo Gagnon, Mizu Nishikawa-Toomey, Sarthak Mittal, Guillaume Lajoie, Dhanya Sridhar

*under review

 Code  PDF

In-Context Learning and Occam's Razor

Eric Elmoznino*, Tom Marty*, Tejas Kasetty, Leo Gagnon, Sarthak Mittal, Mahan Fathi, Dhanya Sridhar, Guillaume Lajoie

International Conference on Machine Learning (ICML 2025)

 Code  PDF

Does Learning the Right Latent Variables Necessarily Improve In-Context Learning?

Sarthak Mittal, Eric Elmoznino, Leo Gagnon, Sangnie Bhardwaj, **Tom Marty**, Dhanya Sridhar, Guillaume Lajoie

International Conference on Machine Learning (ICML 2025)

 PDF

Next-Token Prediction Should be Ambiguity-Sensitive: A Meta-Learning Perspective

Leo Gagnon, Eric Elmoznino, Sarthak Mittal, **Tom Marty**, Tejas Kasetty, Dhanya Sridhar, Guillaume Lajoie

International Conference on Machine Learning (FoMo@ICML 2025)

 PDF

The BrowserGym Ecosystem for Web Agent Research

Thibault Le Sellier De Chezelles, Maxime Gasse, Alexandre Drouin, Massimo Caccia, Léo Boisvert, Megh Thakkar, **Tom Marty**, Rim Assouel, Sahar Omid Shayan, Lawrence Keunho Jang, Xing Han Lù, Ori Yoran, Dehan Kong, Frank F. Xu, Siva Reddy, Quentin Cappart, Graham Neubig, Ruslan Salakhutdinov, Nicolas Chapados, Alexandre Lacoste

Transactions on Machine Learning Research (TMLR 2025) .

 Code  PDF

WorkArena: How Capable Are Web Agents at Solving Common Knowledge Work Tasks?

Alexandre Drouin, Maxime Gasse, Massimo Caccia, Issam H Laradji, Manuel Del Verne, **Tom Marty**, Léo Boisvert, Megh Thakkar, Quentin Cappart, David Vazquez, Nicolas Chapados, Alexandre Lacoste

International Conference on Machine Learning (ICML 2024) .

 Code  PDF

The Unsolved Challenges of LLMs as Generalist Web Agents: A Case Study

Rim Assouel*, **Tom Marty***, Massimo Caccia, Issam H. Laradji, Alexandre Drouin, Sai Rajeswar, Hector Palacios, Quentin Cappart, David Vazquez, Nicolas Chapados, Maxime Gasse, Alexandre Lacoste

Neural Information Processing Systems (FMDM@NeurIPS 2024) .

 Code  PDF

Learning and Fine-Tuning a Generic Value-Selection Heuristic Inside a Constraint Programming Solver

Tom Marty*, Léo Bois-Vert*, Tristan François, Pierre Tessier, Louis Gautier, Léo-Boisvert, Louis-Martin Rousseau, Quentin Cappart, Constraint **Journal** 2024.

 Code  PDF

Learning a Generic Value-Selection Heuristic Inside a Constraint Programming Solver

Tom Marty*, Tristan François, Pierre Tessier, Louis Gautier, Louis-Martin Rousseau, Quentin Cappart

Distinguished paper, Constraint Programming (CP 2023).

 Code  PDF

You can also find the most up-to-date publications on my Google Scholar page.

OTHER PROJECTS

Discrete flow matching for protein co-design aimed at pMHC:TCR binding

Ongoing Research

- Engineered a large-scale distributed training pipeline to train a sequence-structure co-design generative model over 200 million protein sequences and 100000 3D structures

Dissection of concept acquisition in protein Language Model

Ongoing Research

- Curated datasets to evaluate the model's acquisition of known biochemical and structural concepts.

Autonomous Drone Swarm Deployment - DGA contest | *Python, PyTorch*

Nov. 2020 – Mar. 2021

- Multi-agent Q-Learning method for deployment optimization
- Density-Based Spatial Clustering for point of interest detection

Realtime 3D Deep Motion Capture | *C++, OpenCV, PyTorch*

Oct. 2020 – Dec. 2020

- Implemented a method of inferring a full character's 3d pose using only a camera as an input
- Inspired by a EECV 2020 research paper to implement the algorithm

Sketch-based Shape Retrieval | *Python, C++, OpenGL*

Sep. 2020 – Dec. 2020

- Implemented a method to find any specific 3d model in a database using a drawing as an input
- Succeeded to faithfully retrieve several simple 3D shapes by using a single drawing given by a user

Visit  my website to know more about me and my projects!

HONORS AND AWARDS

FRQNT doctoral training scholarship: 100000\$	Mar. 2025
Distinguished Paper Award at CP conference	Sep. 2023
MITACS Accelerate scholarship: 30000\$	Mar. 2023
Outstanding Investment Mention, Ecole Polytechnique	Jul. 2022
Vallet Fondation scholarship: 10000€	Sep. 2018

TEACHING EXPERIENCE

Teaching Assistant	Fall 2025
<i>IFT6390: Fundamentals of machine learning</i>	
Teaching Assistant	Fall 2022
<i>INF8215, Artificial Intelligence : Algorithms and methods</i>	
Teaching Assistant	Fall 2021
<i>INF8215, Artificial Intelligence : Algorithms and methods</i>	
Teaching Assistant	Fall 2018 – Winter 2019
<i>Ministry of National Education</i>	
• Responsible for a group of up to 20 undergraduate students during scientific workshops • Worked alongside the academic team to prepare students for entrance exams	

COMMUNITY INVOLVEMENT AND SERVICE

Tea Talks Committee Member – Research Seminar held at Mila	2025
Reviewer for NeurIPS 2025 NewInML Workshop	2025
Reviewer for NeurIPS 2024 CALM Workshop, MAIS 2024, HRAIM 2024	2024
Reviewer for CP2023	2023
President of Nuit du Styx festival	2020
• Responsible of the organization of an electronic music festival gathering more than 2000 persons	
Member of Rethorix, Public Speaking Club	2019

SKILLS & HOBBIES

Languages: French : Native | English : Fluent
Developer Toolbox: Pytorch, Lightning, Git, Hydra, WandB, SLURM, HPC, CI Testing
Programming Languages: Python, Julia, C++, R
Activities: Climbing, Hiking, Surfing, Snowboarding, UAV robotics

REFERENCE

Prof. Dhanya Sridhar (Ph.D. advisor)
 Assistant Professor at UdeM, Core academic member at MILA - AI CIFAR Chair holder
 Email : dhanya.sridhar@mila.quebec

Dr. Alexandre Lacoste
 Staff Research Engineer, ServiceNow Research
 Email : alexandre.lacoste@servicenow.com

Prof. Quentin Cappart (M.Sc. advisor)
 Assistant Professor at Polytechnique Montréal
 Email : quentin.cappart@polymtl.ca